MSG No. 189.221 $P\bar{6}2m$ [Type I, hexagonal]

Table 1: Wyckoff site: `1a`, site symmetry: `-62m`

No.	position	mapping
1	[0, 0, 0]	`[1,2,3,4,5,6,7,8,9,10,11,12]`

Table 2: Wyckoff site: `1b`, site symmetry: `-62m`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12]`

Table 3: Wyckoff site: `2c`, site symmetry: `-6..`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	`[1,2,3,7,8,9]`
2	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	`[4,5,6,10,11,12]`

Table 4: Wyckoff site: `2d`, site symmetry: `-6..`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	`[1,2,3,7,8,9]`
2	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$	`[4,5,6,10,11,12]`

Table 5: Wyckoff site: `2e`, site symmetry: `3.m`

No.	position	mapping
1	$[0, 0, z]$	`[1,2,3,10,11,12]`
2	$[0, 0, -z]$	`[4,5,6,7,8,9]`

Table 6: Wyckoff site: `3f`, site symmetry: `m2m`

No.	position	mapping
1	$[x, 0, 0]$	`[1,4,8,11]`
2	$[0, x, 0]$	`[2,5,9,12]`
3	$[-x, -x, 0]$	`[3,6,7,10]`

Table 7: Wyckoff site: `3g`, site symmetry: `m2m`

No.	position	mapping
1	$\left[x, 0, \frac{1}{2}\right]$	`[1,4,8,11]`
2	$\left[0, x, \frac{1}{2}\right]$	`[2,5,9,12]`

3	$\left[-x,\, -x,\, \frac{1}{2}\right]$	`[3,6,7,10]`

Table 8: Wyckoff site: `4h`, site symmetry: `3..`

No.	position	mapping
1	$\left[\frac{1}{3},\, \frac{2}{3},\, z\right]$	`[1,2,3]`
2	$\left[\frac{2}{3},\, \frac{1}{3},\, -z\right]$	`[4,5,6]`
3	$\left[\frac{1}{3},\, \frac{2}{3},\, -z\right]$	`[7,8,9]`
4	$\left[\frac{2}{3},\, \frac{1}{3},\, z\right]$	`[10,11,12]`

Table 9: Wyckoff site: `6i`, site symmetry: `..m`

No.	position	mapping
1	$[x,\, 0,\, z]$	`[1,11]`
2	$[0,\, x,\, z]$	`[2,12]`
3	$[-x,\, -x,\, z]$	`[3,10]`
4	$[x,\, 0,\, -z]$	`[4,8]`
5	$[0,\, x,\, -z]$	`[5,9]`
6	$[-x,\, -x,\, -z]$	`[6,7]`

Table 10: Wyckoff site: `6j`, site symmetry: `m..`

No.	position	mapping
1	$[x,\, y,\, 0]$	`[1,8]`
2	$[-y,\, x-y,\, 0]$	`[2,9]`
3	$[-x+y,\, -x,\, 0]$	`[3,7]`
4	$[x-y,\, -y,\, 0]$	`[4,11]`
5	$[y,\, x,\, 0]$	`[5,12]`
6	$[-x,\, -x+y,\, 0]$	`[6,10]`

Table 11: Wyckoff site: `6k`, site symmetry: `m..`

No.	position	mapping
1	$\left[x,\, y,\, \frac{1}{2}\right]$	`[1,8]`
2	$\left[-y,\, x-y,\, \frac{1}{2}\right]$	`[2,9]`
3	$\left[-x+y,\, -x,\, \frac{1}{2}\right]$	`[3,7]`
4	$\left[x-y,\, -y,\, \frac{1}{2}\right]$	`[4,11]`
5	$\left[y,\, x,\, \frac{1}{2}\right]$	`[5,12]`
6	$\left[-x,\, -x+y,\, \frac{1}{2}\right]$	`[6,10]`

Table 12: Wyckoff site: `12l`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$[-y,\ x-y,\ z]$	`[2]`
3	$[-x+y,\ -x,\ z]$	`[3]`
4	$[x-y,\ -y,\ -z]$	`[4]`
5	$[y,\ x,\ -z]$	`[5]`
6	$[-x,\ -x+y,\ -z]$	`[6]`
7	$[-x+y,\ -x,\ -z]$	`[7]`
8	$[x,\ y,\ -z]$	`[8]`
9	$[-y,\ x-y,\ -z]$	`[9]`
10	$[-x,\ -x+y,\ z]$	`[10]`
11	$[x-y,\ -y,\ z]$	`[11]`
12	$[y,\ x,\ z]$	`[12]`